

4(a)	<i>any four from:</i> carbaminohaemoglobin ; A combined with haemoglobin hydrogencarbonate ions / HCO_3^- ; R HCO_3 CO_2 dissolved in the plasma ;	2
4(b)	<i>any four from:</i> - correct direction of movement for oxygen <u>and</u> carbon dioxide ; e.g. oxygen, moves from, alveolus / W, to, capillary / blood / red blood cells, <u>and</u> carbon dioxide moves from, capillary/ blood /plasma to alveolus / blood to W ; - diffusion of, oxygen / carbon dioxide / respiratory gases ; A movement down, partial pressure / concentration, gradient A from high(er), partial pressure / concentration, to low(er), partial pressure / concentration I along a concentration gradient - use of data from Table 4.1 to support movement of oxygen or carbon dioxide ; <i>needs ref. to kPa once</i> <i>allow ecf from mp2</i> (diffusion) through, alveolar wall / squamous epithelium ; (diffusion) through, endothelium / capillary wall ; - AVP ; e.g. haemoglobin binding oxygen	4
4(c)	<i>any four from:</i> alveoli / alveolar walls, stretch / expand, during, inspiration / AW ; arteries / arterioles / blood vessels, stretch / expand, as blood, volume / pressure, increases ; stretch to prevent (alveoli / arteries), bursting / rupture / AW ; A prevent overstretching recoil during expiration / AW, to help, expel / force out, air ; AW <i>in context of alveoli</i> recoil to apply pressure to blood so maintaining blood pressure ; <i>in context of arteries</i> AVP ; e.g. allows trachea / bronchus / bronchioles, to expand during inspiration	4

Question	Answer	Marks
4(d)(i)	$\frac{50}{86} \times 100 ;$ 58(%) ; <i>if extracted values are different allow ecf for calculated answer</i> <i>accept ± 1 mm when reading from graph (49 to 51 and 85 to 87)</i>	2

Question	Answer	Marks
4(d)(ii)	<i>allow 'volume of blood' for 'mean volume of blood per kg body mass'</i> <i>any four from:</i> <i>description to max 3</i> 1. trend is / overall / AW, increase in volume of blood, over time / AW <i>or</i> increase in volume of blood over time except between 0 and 1–2 months ; 2. trend is / overall / AW, increase in red blood cell volume, over time / AW ; <i>or</i> increase in red cell volume over time except between 7–8 and 12 months ; 3. <i>ref. to</i> percentage of mean blood volume represented by red blood cells / haematocrit ; e.g. fluctuates / mainly increasing / decreases between 1–2 and 3–4 months / decreases between 7–8 and 12 months ; 4. any comparative use of, figures / percentages ; <i>to support mp1, mp2 or mp3</i> <i>explanation to max 3</i> 5. lower partial pressure of oxygen at high altitude / less oxygen in inhaled air ; 6. lower oxygen saturation of haemoglobin / haemoglobin has lower oxygen affinity ; 7. compensation <i>or</i> <i>idea that</i> changes occur to make up for less, oxygen taken in / transported to tissues R <i>idea of</i> body getting more oxygen than at sea level 8. (over time) more red blood cells produced so more haemoglobin / AW ; 9. (over time) proportion of red cells / concentration of haemoglobin, increases so, take up / AW, greater volume of oxygen ; A more red blood cells per unit time passes through, lungs to take up oxygen / tissues to deliver oxygen 10. AVP ; e.g. <i>ref. to</i> erythropoietin / EPO	4